

ESPE

UNIVERSIDAD DE LAS FUERZAS ARMADAS

INNOVACIÓN PARA LA EXCELENCIA

University of the Armed Forces

DEPARTMENT OF COMPUTER SCIENCE

Object-Oriented Programming

Edison Lascano PhD

Pachacama Andres

Parra Carlos

Paucar Matias

NRC: 36862

Project's Name: Prov&Tec Management: Comprehensive Sales and Inventory System

Team Name: Byte Works

Problem

Currently, Prov&Tec manages its **sales** and **inventory** entirely manually, relying on paper **documents** (**sales slips** and **order forms**) and basic **spreadsheets**. Lacking technological **tools**, the **company** has no precise daily control over its **inventory** of **electrical materials**.

The main bottleneck is in the **warehouse**, where the recording of incoming and outgoing **goods** is slow and inconsistent. This lack of synchronization and reliable **information** leads to logistical disarray: **staff** waste excessive time searching for misplaced **products**, and-unaware of actual **stock** levels the **company** orders **materials** it already has on hand, resulting in unnecessary expenses.

Consequently, **customer** service suffers significantly. **Buyers** wait 10 to 20 minutes while **salespeople** physically search for less common **items** and check **paperwork**; this causes dissatisfaction and leads many **customers** to leave the **store** without completing their **purchase**. Compounding the issue is an accumulation of slow-moving **stock**, which generates significant financial losses and limits the **owner**'s management capacity. Therefore, implementing a computer **system** to centralize and automate these **processes** is essential.

Overview

The Sales and **Inventory Management System** for Prov&Tec is an object-oriented software **solution** designed to centralize and automate the logistical and commercial management of **electrical material sales**. Its primary purpose is to replace manual **methods** and paper-based **documentation** with a digital **platform** that minimizes errors and provides reliable **data** for daily **operations**.

The **system** will feature secure access with defined **user roles** (such as **salesperson** or **warehouse manager**). Its **modules** will allow for the recording and updating of **customer**, **product**, and **supplier information**. To streamline **customer** service, **salespeople** will be able to quickly process **sales**, while the **system** updates **inventory** levels in real time with every **sale** or **supplier purchase**.

Finally, to give the **owner** greater control over the **business**, the **system** will include a **transaction history**, a **reporting module**, and an **alert system** that automatically notifies the **user** when **products** reach minimum **stock** levels. This helps prevent stockouts and facilitates strategic decision-making.

Background

Since its inception, Prov&Tec has operated using a strictly manual management **model**. Although the **company** has managed to sustain itself, grow, and overcome significant obstacles such as financial losses due to stagnant **i**nventory that failed to turn over-its reliance on paper limits its efficiency. Consequently, the company seeks to make a technological leap to modernize its logistics **processes**, maintain rigorous **inventory** control, and offer more agile service.

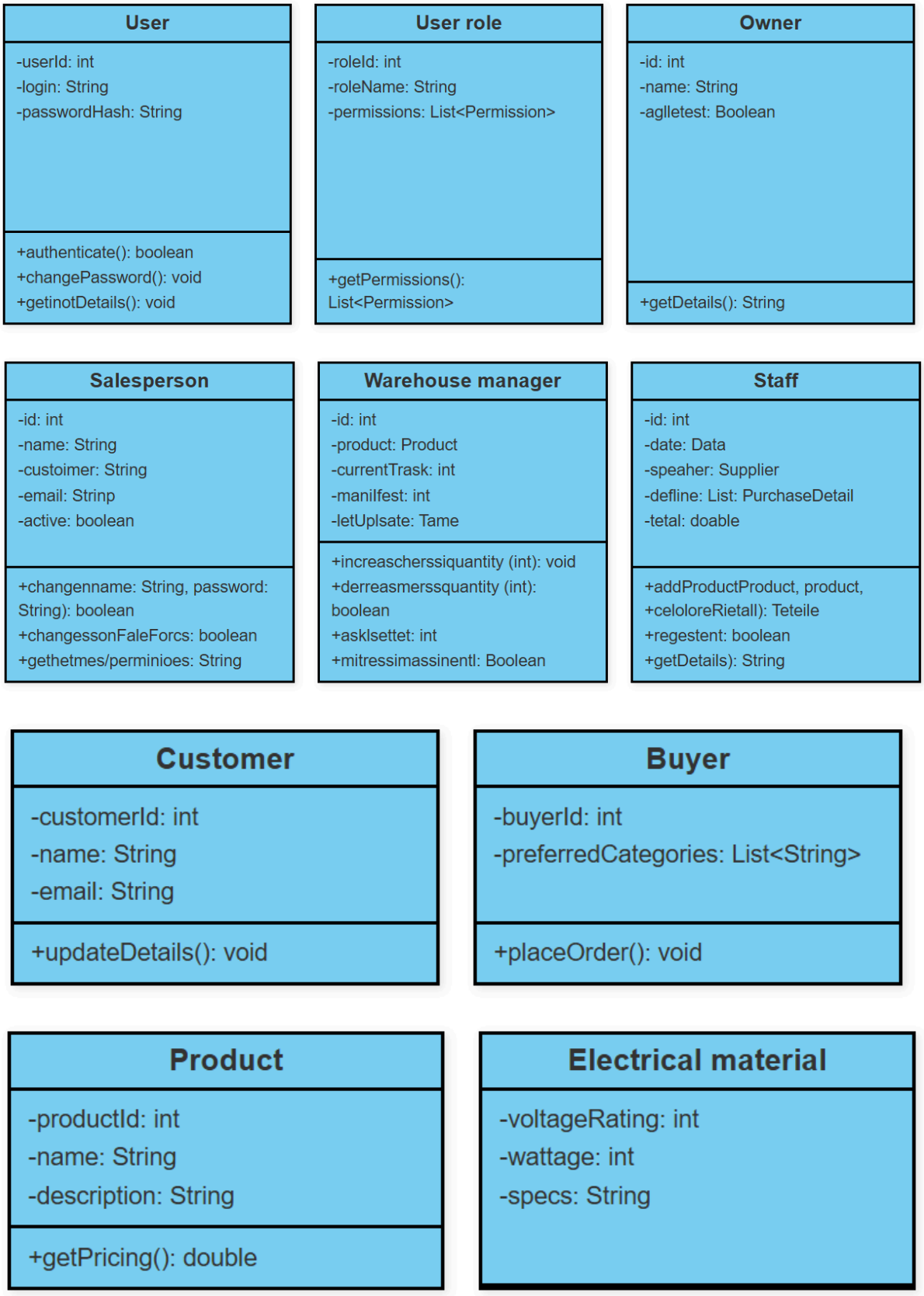
Compounding this situation is the market's digital transition, as traditional **processes** such as physical **invoicing** rapidly shift toward electronic **formats**. The need to reduce the excessive administrative **paperwork** burdening the owner, Ximena Añamise, presented a clear opportunity for a technical solution.

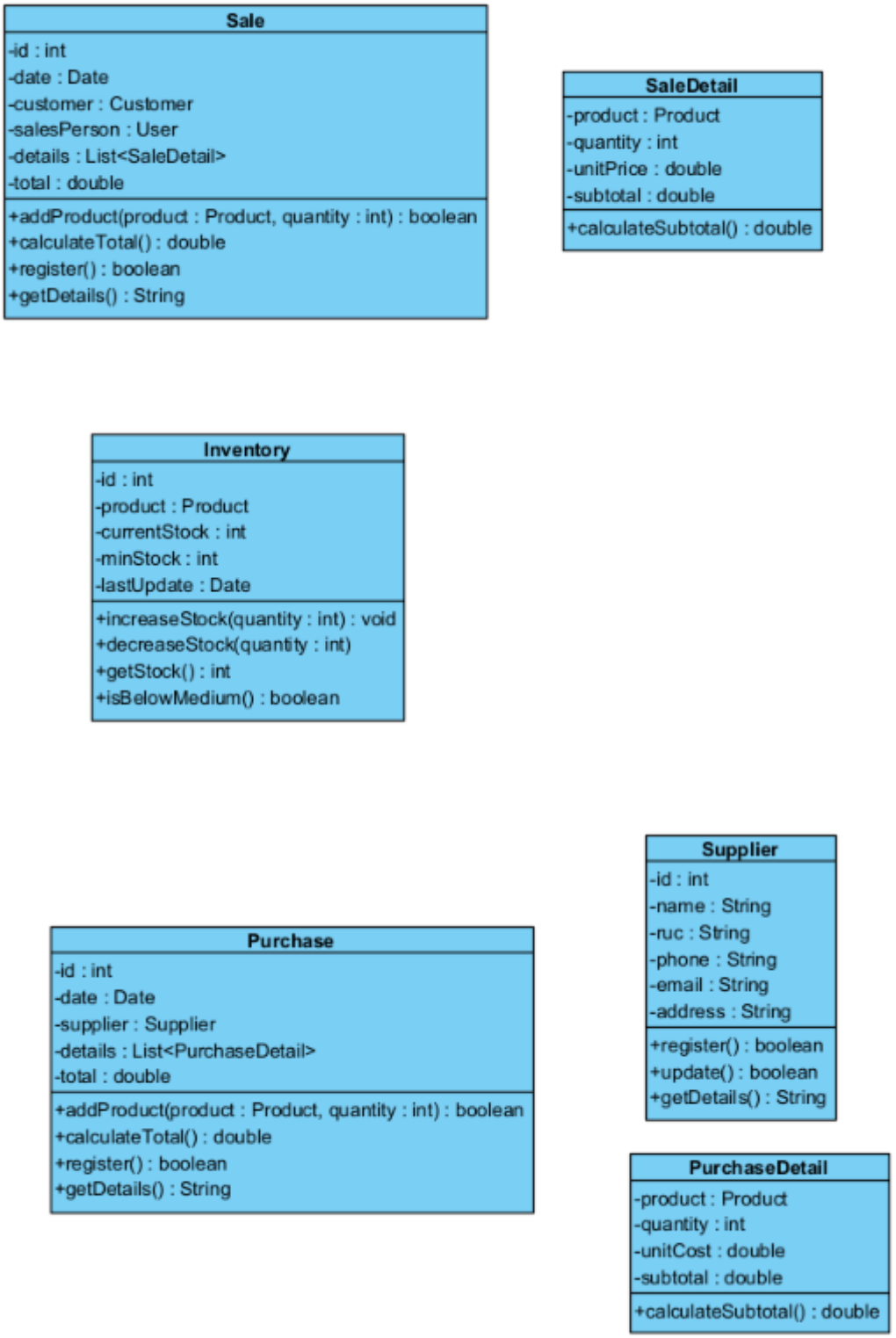
As Object-Oriented Programming students at the University of the Armed Forces (ESPE) and aspiring software engineers, the development team decided to apply their knowledge to solve this real-world problem. This project allows them to put class and object modeling into practice in a production environment while providing Prov&Tec with a tangible tool to enter the digital age, centralize its data, and streamline its daily operations.

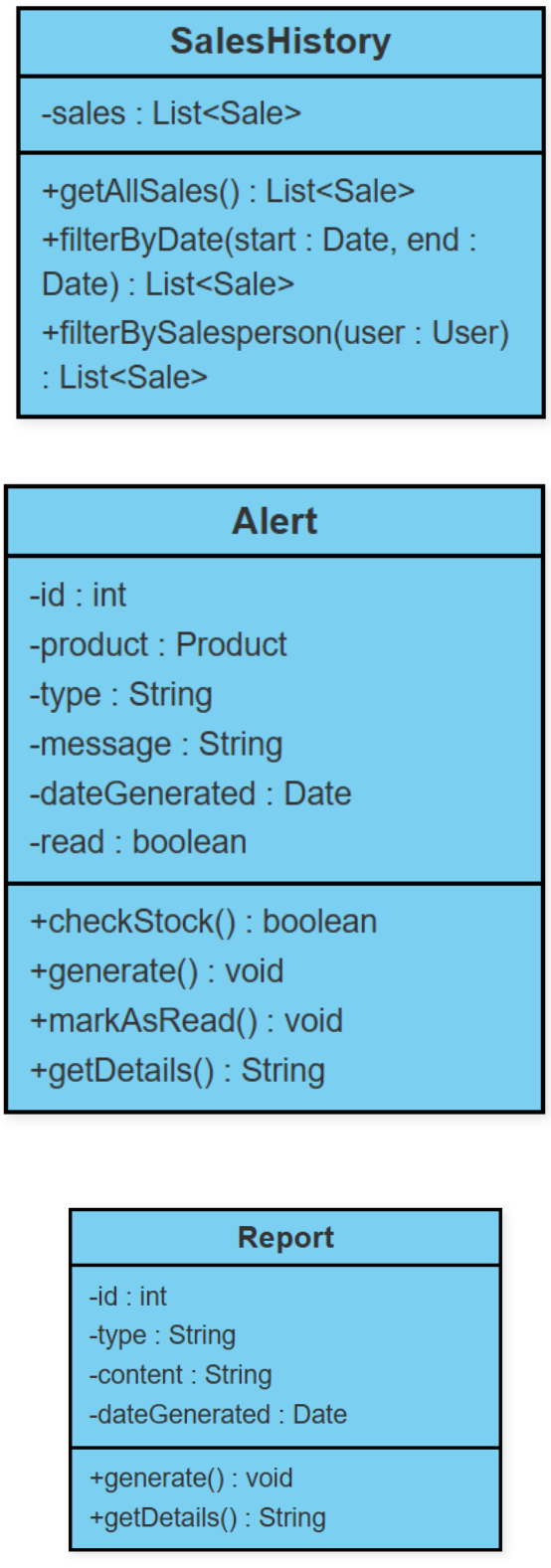
- **Nouns**

Priority: High
Done: No

User	Alert system
Role	Business
Customer	Buyer
Product	Company
Sale	Data
Sale Detail	Document
Stock	Electrical material
Supplier	Engineer
Purchase	Format
Sales history	Method
Alert	Modules
Report	Operations
Order forms	Processes
Spreadsheets	Stockouts
Students	Warehouse
Staff	Purchase
Paperwork	Platform


User
-userId: int
-login: String
-passwordHash: String
+authenticate(): boolean
+changePassword(): void
+getinotDetails(): void
User role
-roleId: int
-roleName: String
-permissions: List<Permission>
+getPermissions(): List<Permission>
Owner
-id: int
-name: String
-aglletest: Boolean
+getDetails(): String
Salesperson
-id: int
-name: String
-custoimer: String
-email: Strinp
-active: boolean
+changenname: String, password: String): boolean
+changessonFaleForcs: boolean
+gethetmes/perminioes: String
Warehouse manager
-id: int
-product: Product
-currentTrask: int
-manilfest: int
-letUplsate: Tame
+increascherssiquantity (int): void
+derreasmerssquantity (int): boolean
+asklsettet: int
+mitressimassinentl: Boolean
Staff
-id: int
-date: Data
-speaher: Supplier
-defline: List: PurchaseDetail
-tetal: doable
+addProductProduct, product,
+celoloreRietall): Teteile
+regestent: boolean
+getDetails): String
Customer
-customerId: int
-name: String
-email: String
+updateDetails(): void
Buyer
-buyerId: int
-preferredCategories: List<String>
+placeOrder(): void
Product
-productId: int
-name: String
-description: String
+getPricing(): double
Electrical material
-voltageRating: int
-wattage: int
-specs: String


Sale
-id : int
-date : Date
-customer : Customer
-salesPerson : User
-details : List<SaleDetail>
-total : double
+addProduct(product : Product, quantity : int) : boolean
+calculateTotal() : double
+register() : boolean
+getDetails() : String
SaleDetail
-product : Product
-quantity : int
-unitPrice : double
-subtotal : double
+calculateSubtotal() : double
Inventory
-id : int
-product : Product
-currentStock : int
-minStock : int
-lastUpdate : Date
+increaseStock(quantity : int) : void
+decreaseStock(quantity : int)
+getStock() : int
+isBelowMedium() : boolean
Supplier
-id : int
-name : String
-ruc : String
-phone : String
-email : String
-address : String
+register() : boolean
+update() : boolean
+getDetails() : String
Purchase
-id : int
-date : Date
-supplier : Supplier
-details : List<PurchaseDetail>
-total : double
+addProduct(product : Product, quantity : int) : boolean
+calculateTotal() : double
+register() : boolean
+getDetails() : String
PurchaseDetail
-product : Product
-quantity : int
-unitCost : double
-subtotal : double
+calculateSubtotal() : double


SalesHistory
-sales : List<Sale>
+getAllSales() : List<Sale>
+filterByDate(start : Date, end : Date) : List<Sale>
+filterBySalesperson(user : User) : List<Sale>
Alert
-id : int
-product : Product
-type : String
-message : String
-dateGenerated : Date
-read : boolean
+checkStock() : boolean
+generate() : void
+markAsRead() : void
+getDetails() : String
Report
-id : int
-type : String
-content : String
-dateGenerated : Date
+generate() : void
+getDetails() : String